

**Step 3.** The MI factor score is then:

$$K(\text{Shares}) = \underbrace{a_1 \cdot \left( \frac{1}{\text{ADV}} \right)^{a_2}}_{\text{MI Factor Score}(\text{Shares})} \cdot \sigma^{a_3} \quad (11.13)$$

The MI factor score provides managers with the stock's market impact sensitivity  $K(\text{Shares})$ . For the same number of shares to trade and the same strategy, the MI factor score will provide a ranking value of each stock's trading cost. This score allows a fair and consistent comparison of cost of trading across stocks and includes both liquidity and volatility terms. If the MI factor score is twice as high for one stock compared to another stock, the cost to trade the first stock will be twice as high as for the second stock.

Portfolio managers finally have a trading cost factor score that will alleviate the need to utilize broker-dealer pre-trade models, and perform time consuming numerical procedures. The factor score can be easily computed on the manager's desktop or integrated into a proprietary in-house model. The only requirement to compute the score is to have the market impact parameters and stock trading characteristics (see Kissell, 2013).

In many situations, portfolio managers are not setting out to invest in a specified number of shares or in a specified order size (% ADV). They are more often setting out to invest a specified dollar value into a stock or basket of stocks. To accommodate these needs we can reformulate the MI factor score in terms of dollars.

The number of shares that can be purchased for a fixed dollar amount is:

$$\text{Shares} = \frac{\text{Dollars\$}}{\text{Price}} \quad (11.14)$$

Then we can compute the MI factor score in terms of dollars as follows:

**Step 1a.** Start with the I-Star model:

$$I^*(\text{Shares}) = a_1 \cdot \left( \frac{\text{Shares}}{\text{ADV}} \right)^{a_2} \cdot \sigma^{a_3} \quad (11.15)$$

**Step 1b.** Convert shares to dollars:

$$I^*(\text{Dollars\$}) = a_1 \cdot \left( \frac{\text{Dollars\$}}{\text{Price}} \cdot \frac{1}{\text{ADV}} \right)^{a_2} \cdot \sigma^{a_3} \quad (11.16)$$